

Fenoprazil Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-295/A

Container Closure System

The commercial packaging configuration consists of high-density polyethylene bottles with child-resistant polypropylene closures and induction seal liners. Container closure integrity was demonstrated by dye ingress testing. The packaging components comply with applicable food-contact regulations and compendial requirements for plastic packaging systems.

The control strategy links critical quality attributes to critical process parameters identified during development. No changes to the approved analytical procedures are proposed in this submission. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. Cleaning verification swab results were below the health-based exposure limits derived for the product. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. The control strategy links critical quality attributes to critical process parameters identified during development. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

The control strategy links critical quality attributes to critical process parameters identified during development. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. The control strategy links critical quality attributes to critical process parameters identified during development. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Deviations observed during the campaign were investigated and closed with no impact to product quality.

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

Statistical evaluation of batch release data demonstrates process capability well within specification limits. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Deviations observed during the campaign were investigated and closed with no impact to product quality. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. No changes to the approved analytical procedures are proposed in this submission. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Deviations observed during the campaign were investigated and closed with no impact to product quality. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Cleaning verification swab results were below the health-based exposure limits derived for the product. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

The control strategy links critical quality attributes to critical process parameters identified during development. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Cleaning verification swab results were below the health-based exposure limits derived for the product.

3.2.P.4 Excipient Control Strategy

For the commercial formulation F-295/A, Methocel K4M Premium CR (Hypromellose, release-controlling polymer) is used.

Grade selection and control follow the principles of IPEC Excipient Qualification Guide, ICH Q8(R2) and ICH Q3D.

Excipient Grade Selection

The basis for selection of Methocel K4M Premium CR is lower moisture content mitigating hydrolytic degradation of the drug substance, demonstrated across three pilot-scale batches.

Approved Drug Product Specification

The current approved commercial specification (SPEC-EX-2346 Rev 06) lists Hypromellose as Methocel K100M Premium CR.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Deviations observed during the campaign were investigated and closed with no impact to product quality.

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Cleaning verification swab results were below the health-based exposure limits derived for the product. Deviations observed during the campaign were investigated and closed with no impact to product quality. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.